

Digital Illegal Money Chain Detection

Bank Transaction Pattern Analysis — Hackathon Research Report

1. Objective

Uploaded bank-statement data me aise behavioural patterns identify karna jo further investigation ke liye useful ho sakte hain. Kisi single transaction ko illegal declare nahi kiya jana chahiye; multiple signals ko combine karke risk indication banana chahiye.

2. Observed Transactions

Date	Amount	Sender
13/09/2026	₹299.30	PRASANT KUMAR SAHOO
13/09/2026	₹99.50	NITISH KARMAKAR
13/09/2026	₹99.90	DEEPAK KUMAR
13/09/2026	₹500.00	Arun
14/09/2026	₹299.30	MAHIDA VIJAYSINH BHUPATSI
14/09/2026	₹199.20	Parvaiz Ahmad Dar
14/09/2026	₹299.40	DEVA RAM RAM
14/09/2026	₹299.20	MOHIT SINGH NEGI

3. Amount Clustering

₹299 ke aas-paas repeated credits clearly visible hain: ₹299.30, ₹299.30, ₹299.40 aur ₹299.20. Ye different senders se same masked receiver account me aaye. Website me ise Amount Cluster Detection ke roop me measure kiya ja sakta hai.

4. Decimal / Non-Round Pattern

₹99.50, ₹99.90, ₹299.30, ₹199.20, ₹299.40 aur ₹299.20 jaise non-round amounts observed hain. System decimal suffix frequency, repeated values aur similar-value transactions ko analyse kare. Decimal amount alone suspicious hone ka proof nahi hai.

5. Multiple Senders → Same Receiver

PRASANT, NITISH, DEEPAK, MAHIDA, PARVAIZ, DEVA RAM aur MOHIT jaise different senders ne same masked account XX9342 ko UPI credits bheje. Isse multi-sender concentration signal nikala ja sakta hai.

6. Time-Window Analysis

10-minute, 30-minute, 1-hour, 6-hour aur 24-hour windows me transaction count, unique senders, total amount aur amount similarity analyse ki ja sakti hai. Short window me repeated multi-sender activity additional risk signal ho sakti hai.

7. Money-Chain Detection

Agar statement me incoming aur outgoing records available hain, to Node = account/merchant aur Edge = transaction ke roop me graph banaya ja sakta hai. Isse $A \rightarrow B \rightarrow C \rightarrow D$ chains, fan-in, fan-out aur possible circular flows identify kiye ja sakte hain.

8. Combined Detection Logic

Same receiver + multiple unique senders + similar amounts + short time window + repeated occurrence ko combined behavioural signal maana ja sakta hai. Multiple independent signals hone par risk score badh sakta hai.

9. Prototype Risk Scoring

Signal	Weight
Repeated decimal / non-round pattern	+10
Repeated transactions	+15
Possible transaction splitting	+20
High multi-sender concentration	+20
Rapid onward transfer	+15
Unusual merchant behaviour	+15
Circular flow	+25

Prototype thresholds: 0–20 Low, 21–40 Medium, 41–60 High, 61+ Critical. Real deployment me thresholds validated data aur domain expertise se calibrate hone chahiye.

10. Website Output

Bank Statement Upload → Data Extraction → Transaction Cleaning → Pattern Detection → Graph/Chain Analysis → Risk Scoring → Explainable Report. Report me account summary, suspicious transactions, amount clusters, decimal patterns, counterparties, time-window anomalies, connected accounts, money trail aur evidence table dikhaya ja sakta hai.

11. Security & Privacy

OTP, UPI PIN, password ya banking login credentials collect nahi kiye jane chahiye. Sensitive identifiers ko mask karke display karna chahiye. Analysis authorized statement data par based hona chahiye.

12. Conclusion

Is sample ka strongest combined signal hai: repeated similar amounts + non-round decimals + multiple senders + same receiver + time clustering. Project ka core principle: transaction ko akela analyse nahi karna, balki transaction ke relationship, sequence aur network ko analyse karna.

Note: Suspicious pattern illegal activity ka proof nahi hota. Final determination appropriate evidence aur authorized investigation par depend karega.